Homework #2: Simple Neural Network with NumPy

Building a 1-layer neural network from scratch using only NumPy.

Step 1: Sigmoid Function and Derivative

The sigmoid activation function:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Its derivative (used in backpropagation):

$$\frac{d\sigma(x)}{dx} = \sigma(x)(1-\sigma(x))$$

Important: The derivative must be evaluated on the *pre-activation* value (before sigmoid), not on the output directly.

Step 2: Weight Initialization

We use a (3×1) weight vector because the input has 3 features and we predict 1 output. Weights are drawn from a normal distribution with mean 0 and std 1/3:

$$w \sim \mathcal{N}(\mu = 0,\ \sigma = \tfrac{1}{3})$$

Step 3: Training Loop

Each iteration performs:

Forward pass:

$$\hat{y} = \sigma(X \cdot w)$$

Error:

$$\delta = y - \hat{y}$$

Gradient (chain rule):

$$\nabla w = X^T \cdot (\delta \odot \sigma'(X \cdot w))$$

Weight update:

$$w \leftarrow w + \nabla w$$

Results after 1000 Iterations

The network correctly learns that x_1 is the relevant feature:

x1	x2	x3	y-hat	y
0	0	1	~0.03	0
0	1	1	~0.03	0
1	0	1	~0.97	1
1	1	1	~0.97	1